

FEW-SHOT LEARNING

Ghulam Mustafa
Daniyal Farooqui

PROBLEM

- Powerful Models like Bert, GPT are pre-trained on the entire internet.
- But what if we want to fine-tune them for a specific task? we don't usually have many labeled examples available.
- Meanwhile we as humans, just need a small amount of data to understand the gist of it (we don't need to see 1000s of images of a dog to learn that it's a dog, one or two images are sufficient).
- How do we get models to learn and generalize from minimal amount of data?

FEW-SHOT LEARNING

- Few-shot learning is a machine learning problem where our model has to make predictions after seeing new data from very small amounts of examples (shots)
- The N-way K-shot pattern
- N: number of classes
- K: examples per class



- We classify a new “query” example based on k “support” examples

Support set



Query set



- Zero-shot learning: $K = 0$ (0 examples provided per class)
- One-shot learning: $K = 1$ (1 example provided per class)
- Few-shot learning: $K = 1-10$ (few examples provided per class)

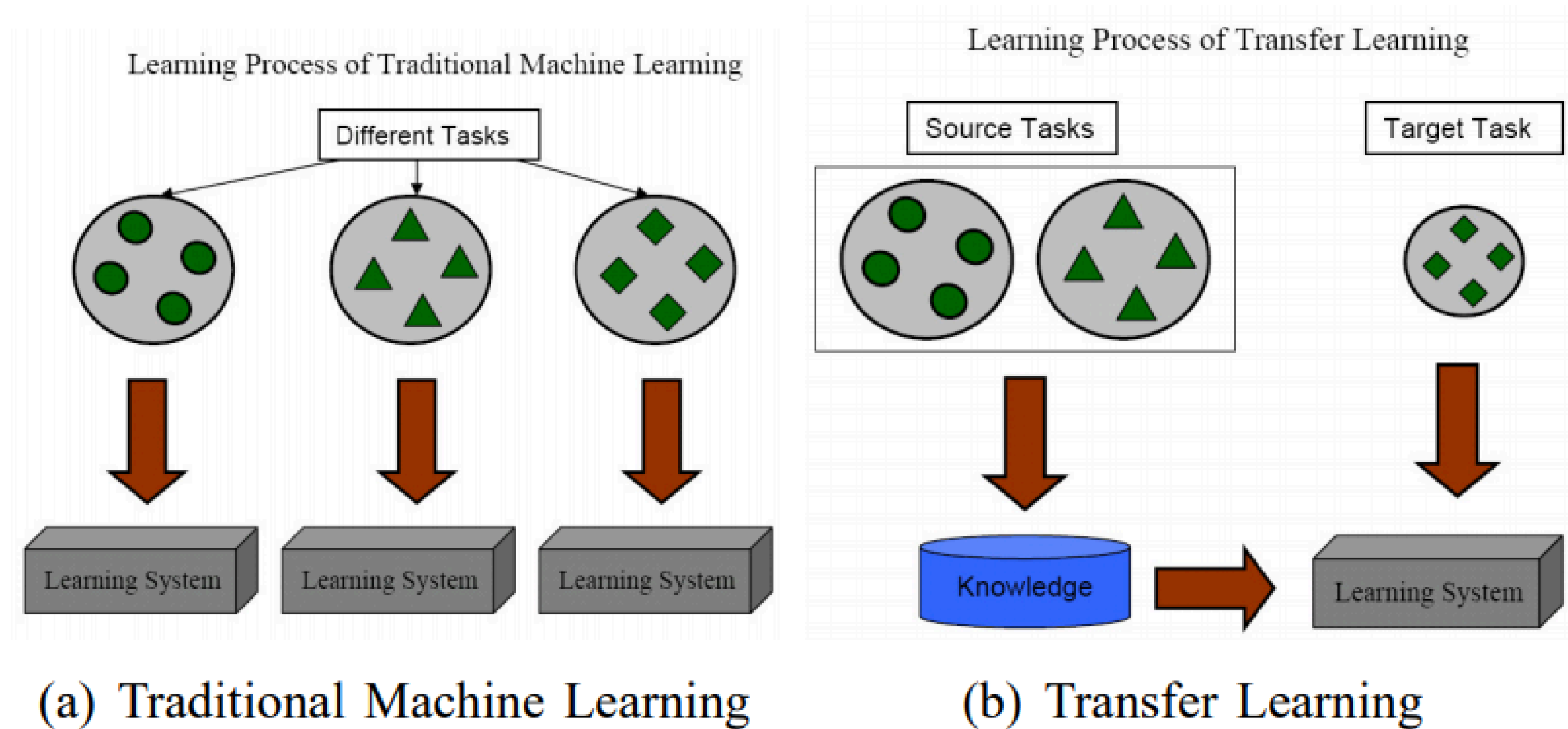
HOW TO SOLVE THIS?

- Traditional vs modern methods
- Traditional approaches include transfer learning and meta learning.
- Modern approaches shifted the focus towards in-context learning

TRANSFER LEARNING

- Methods focused on Transfer Learning involve fine-tuning a pre-trained model to learn new tasks
- Instead of re-training from scratch, we re-train the outer layers while maintaining the inner layers where Feature extraction takes place
- More better results when model's intial training is relevant to the new task (a model trained on a certain species of birds will generalize well to unsees species of birds)

TRANSFER LEARNING



TRANSFER LEARNING

- Source Task: Classifying handwritten digits (MNIST) or (ImageNet).
- Target Task: Detecting fabric defects.
- Method: You are "transferring" the weights learned from the first task to solve the second task.

META LEARNING

- Methods focused on Meta Learning focus on training models from scratch.
- We train the models to be good at learning new tasks
- Making the model train for 1000's of tasks allows it to learn the general patterns of learning.
- Eg: we regularly ask our model how similar are two pictures, over time it becomes an expert on finding similarity since it knows on what features to focus. When we then give it a new pic and ask whether its similar to a hippo or zebra, model will answer perfectly since its an expert on similarity (despite never having seen a hippo or a zebra)

META LEARNING

- We are training a model to detect Sentiment (Positive vs. Negative). This is a 2-Way, 2-Shot Task.
 - 2-Way: 2 Classes (Positive, Negative).
 - 2-Shot: 2 Examples per class to learn from
- **1. The Task (T_i)**
 - A single "Task" is a specific domain or topic we want to master for a moment.
 - Task 1: Sentiment in Movie Reviews.
 - Task 2: Sentiment in Tweets.
 - Task 3: Sentiment in Medical Reports.

META LEARNING

- **The Support Set (S)** ---> "The Study Material"
- The model uses these labeled examples to update its weights.

Class	Text Input (Review)	Label
Positive	"The plot was absolutely brilliant!"	1
Positive	"I loved every second of it."	1
Negative	"This was a total waste of money."	0
Negative	"Boring script and bad acting."	0

META LEARNING

- **The Query Set (Q)** ---> "The Test"
- The model must predict the label for this new sentence to see if it learned.

Text Input (Review)	Expected Label
"I fell asleep halfway through."	? (Target: 0)

META LEARNING

MODEL-AGNOSTIC META-LEARNING (MAML)

META LEARNING

Theta are parameters of the meta model (that's what we are learning).

Loss function when we pass the support data through the meta model $F(\theta)$

Theta' shows the parameters after being updated by the gradient descent.

The diagram illustrates the MAML equation with several annotations:

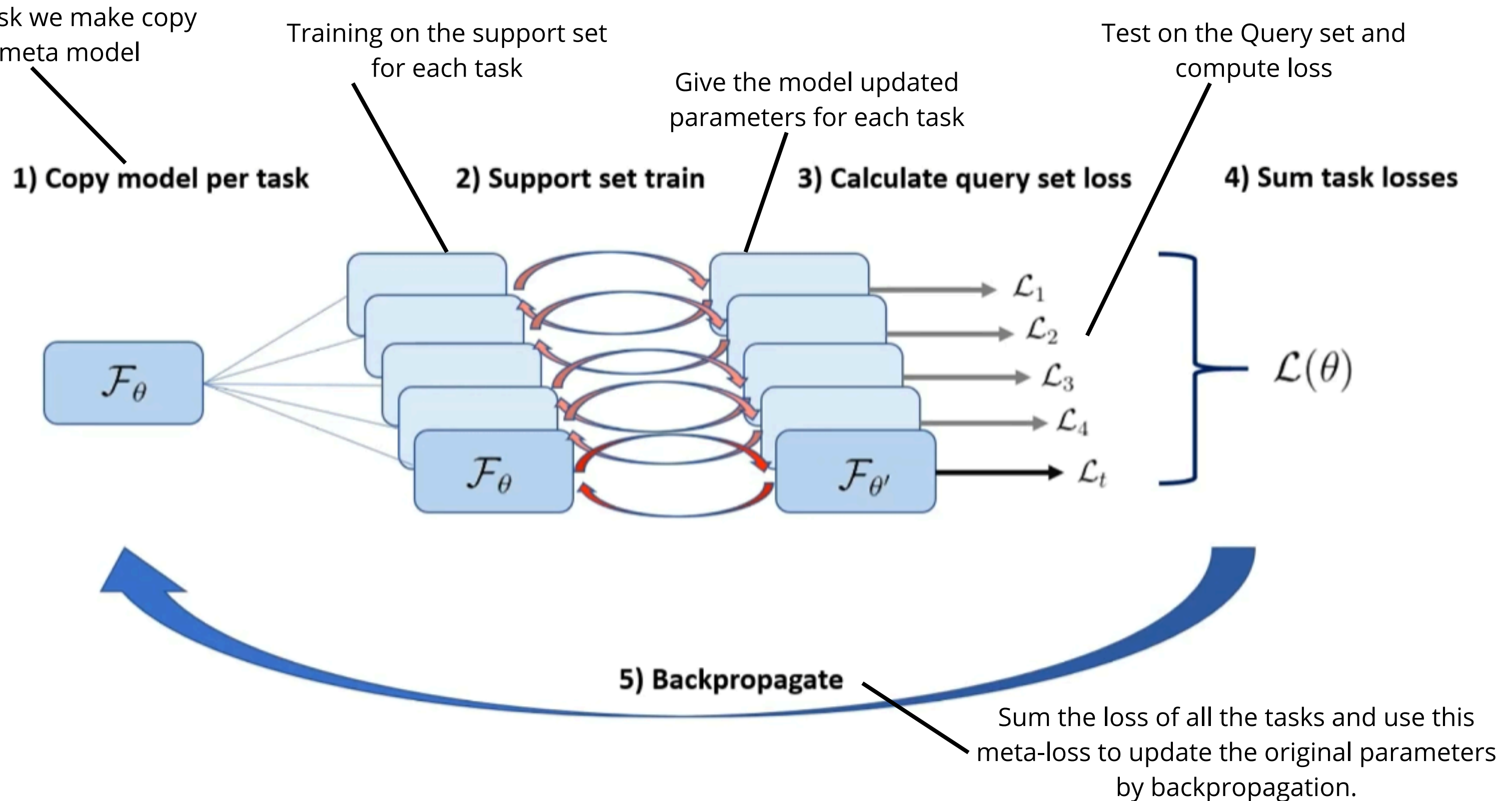
- f_θ Meta Model: Points to the θ in the first equation.
- Loss function: Points to the $\mathcal{L}(\theta - \alpha \nabla_\theta \mathcal{L}(\theta, \mathcal{D}_{\mathcal{T}_i}^S), \mathcal{D}_{\mathcal{T}_i}^Q)$ term.
- Support Set: Points to $\mathcal{D}_{\mathcal{T}_i}^S$.
- Query Set: Points to $\mathcal{D}_{\mathcal{T}_i}^Q$.
- Updated Parameters: Points to θ' .
- Gradient Descent: Points to the $\alpha \nabla_\theta$ term.
- Tasks: Points to the summation over $\mathcal{T}_i \sim p(\mathcal{T})$.

$$\min_{\theta} \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}(\theta - \alpha \nabla_{\theta} \mathcal{L}(\theta, \mathcal{D}_{\mathcal{T}_i}^S), \mathcal{D}_{\mathcal{T}_i}^Q) = \min_{\theta} \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}(\theta', \mathcal{D}_{\mathcal{T}_i}^Q)$$

Summing over a larger number of tasks

So essentially MAML train for few-shot generalization by optimizing for a set of parameters θ such that one or a few steps of gradient descent on the support set achieves a good performance on the query set data

META LEARNING



IN-CONTEXT LEARNING

- Refers to the ability of a large language model to learn a new task instantly by looking at the examples in the prompt, without any update to training or weights
- Source: Brown et al. (2020).
- Used gpt-3 (10x larger than previous state-of-the-art model T5-11B)
- Hypothesis: If the parameters of model are scaled up enough, then the model's "in-context" learning abilities will emerge, hence no need for fine tuning

IN-CONTEXT LEARNING VS TRADITIONAL FINE-TUNING

The three settings we explore for in-context learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



HOW IS IN-CONTEXT LEARNING HAPPENING

- GPT-3 could learn without weight updates. Since we can't "read the mind" of a neural network easily, researchers came up with these two mathematical hypotheses to explain what is happening inside the layers.
- **The Bayesian Framework for In-Context Learning**
- **LLMs are Implementing Learning Algorithms (Implicit Gradient Descent)**

THE BAYESIAN FRAMEWORK FOR IN-CONTEXT LEARNING

- **The Core Hypothesis:**
 - The model does not learn new skills from the prompt.
 - Instead, it uses the prompt to "locate" a specific concept it already learned during pre-training.
- The model implicitly performs Bayesian inference:

$$P(\text{Task} \mid \text{Prompt})$$

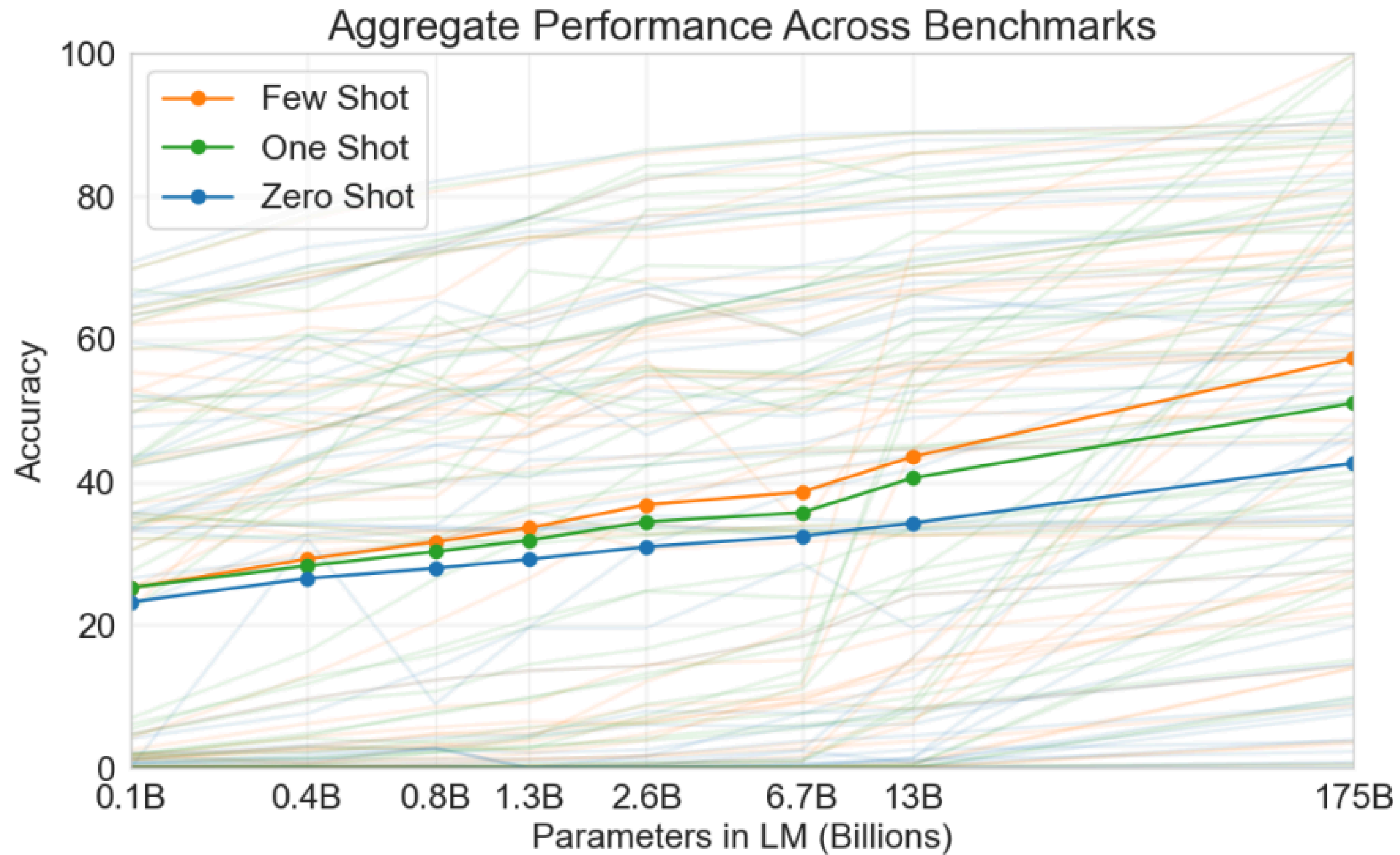
- **Prior:** The model's vast knowledge base.
- **Posterior:** The specific task "activated" by the examples.

- Under this theory, the examples like 'Pakistan: Islamabad, Japan: Tokyo' are not teaching the model what a capital is; it already knows what capitals are.
- Instead, these examples act as evidence or coordinates.
- They tell the model, '**Go to the Language section, find the capital knowledge, and use that.**'
- The model uses Bayesian inference to calculate which concept fits the prompt best, essentially '**locating**' the correct skill without changing a single weight.

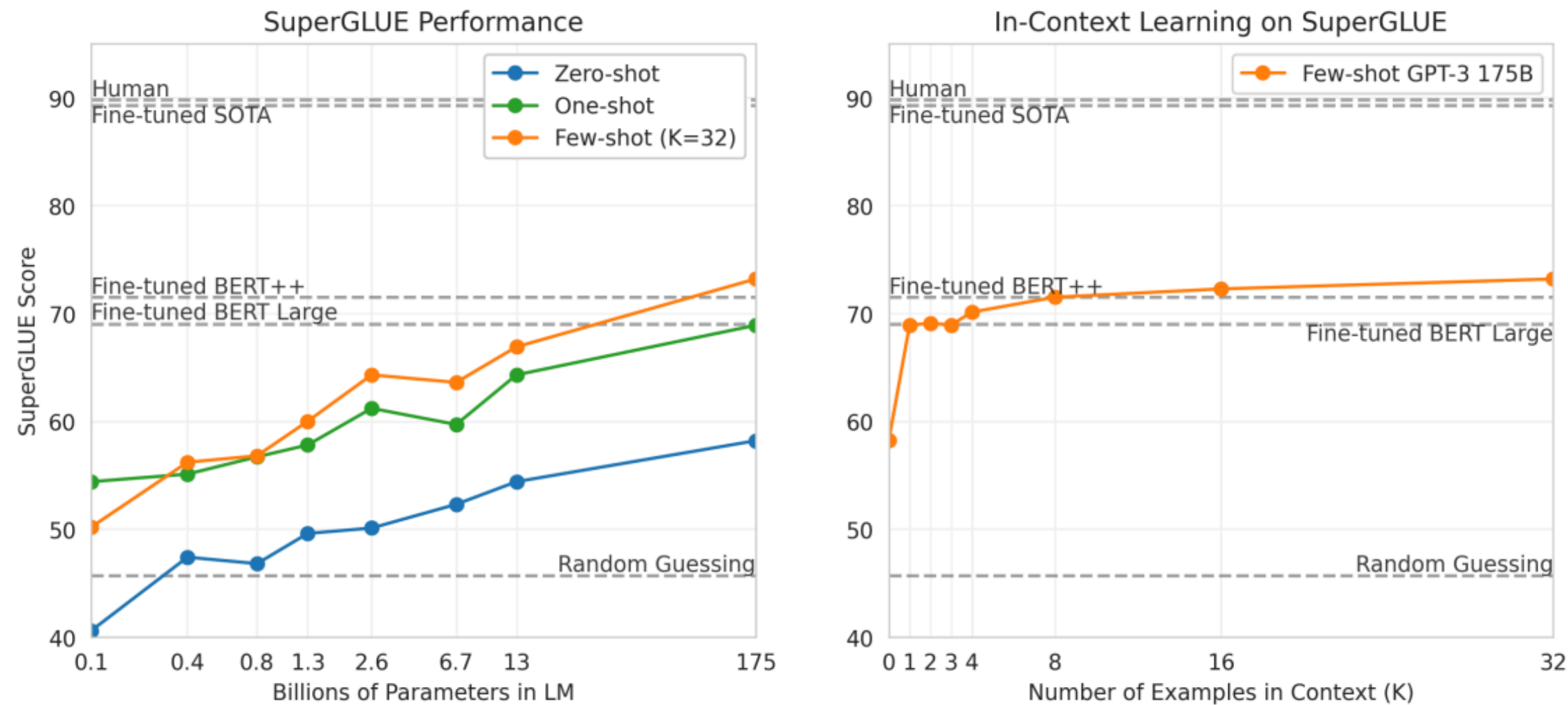
LLMS ARE IMPLEMENTING LEARNING ALGORITHMS (IMPLICIT GRADIENT DESCENT)

- **The Core Hypothesis:**
 - LLMs are not just "looking up" answers; they are actively computing a better model on the fly.
 - The model "simulates" the process of fine-tuning inside its hidden states during the forward pass.
- This means that when you feed examples into GPT, it is essentially running a 'virtual' fine-tuning session. It's not changing its permanent weights, but it is updating its temporary 'activations' to minimize error on your examples.
- **Standard Learning:** Update Weights (W) ---> Permanent Change.
- **In-Context Learning:** Update Activations (A) ---> Temporary Change.

DOES IT WORK?

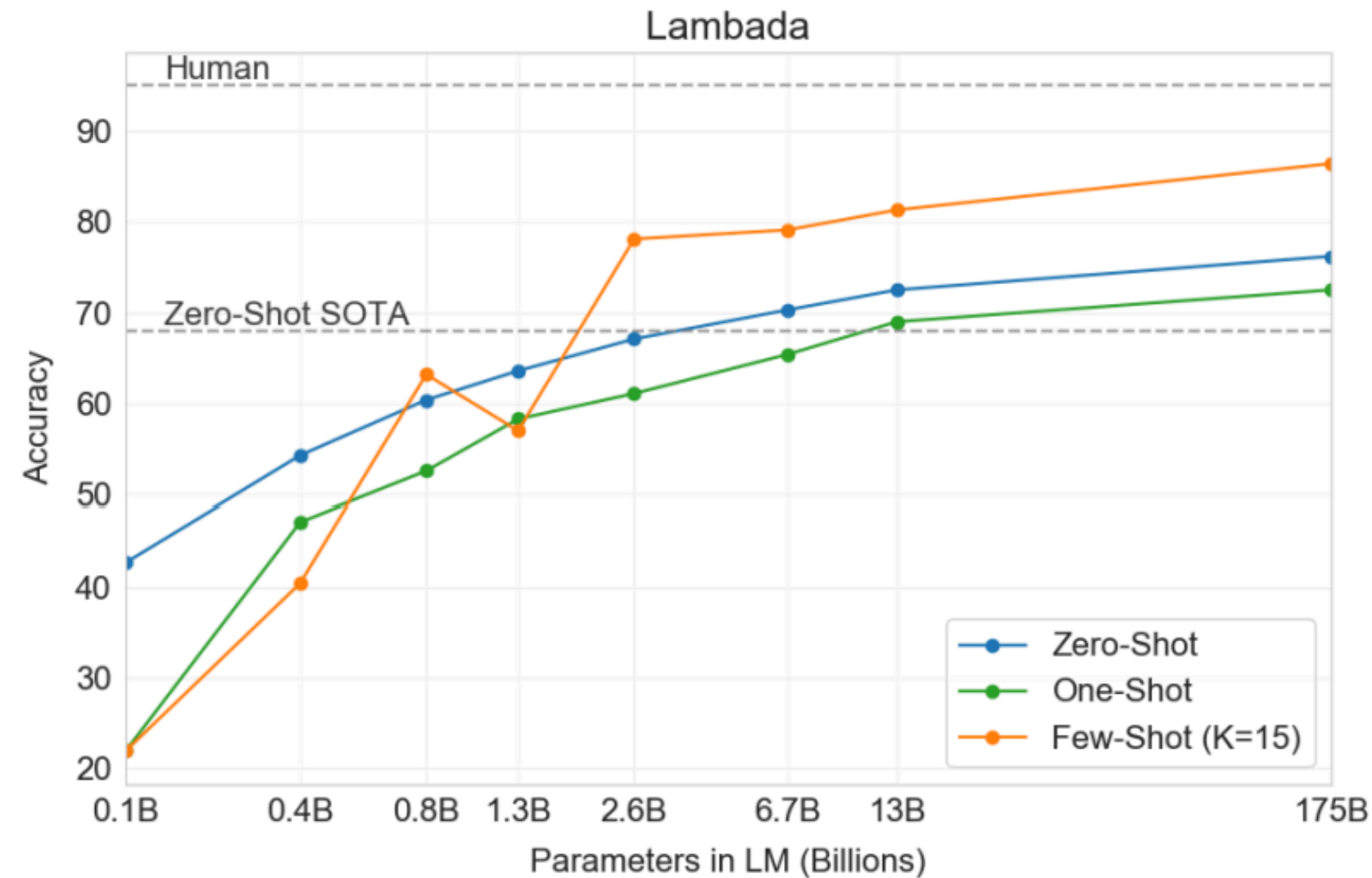


CAPABILITIES



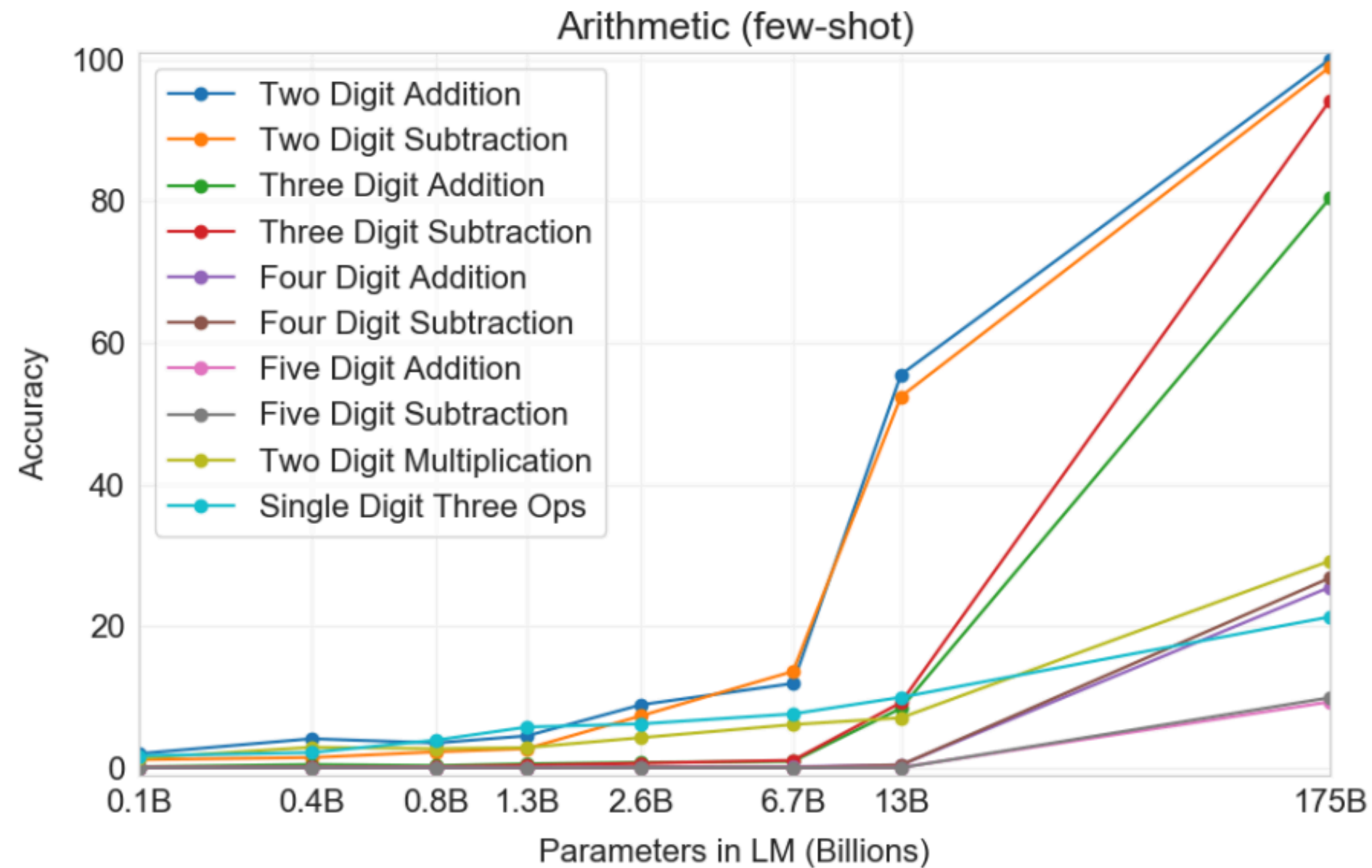
- superGLUE: A collection of difficult language tasks, harder for computers.
- BERT models were specifically fine tuned on superGLUE training set (depicted by dotted lines) unlike GPT-3.
- Left Graph: GPT-3 (orange line) crosses BERT models in superGLUE score
- Right Graph: Giving more number of examples (shots) helps.

CAPABILITIES



- LAMBADA: Test for long-range dependencies. Model reads the whole paragraph and has to predict the missing word in the last sentence.
- GPT-3 2.7B outperforms Zero-shot SOTA 17B
- Shows that model isn't blindly predicting, instead it's tracking the story over long distances.

CAPABILITIES



- Significant jump from GPT-13B to GPT-175B
- Shows that model can learn rules/algorithms (like basic arithmetic) from text patterns alone

- Hence by 2020, we knew that giant models like GPT-3 could solve tasks just by being asked in the prompt.
- BUT: Running a 175B parameter for every single user query is expansive, and GPT-3 struggled with rare tasks like detecting logical fallacies (even with few examples).
- This led to the next evolution in 2024, where instead of using the LLM as the worker (make it do tasks), we use it as a teacher (by generating data for other models to train).

LLMS AS DATA GENERATORS

- Source: Alhindi et al. (ACL 2024)
- Problem: Real world data is imbalanced, many “easy” examples available but few “rare” examples.
- Researchers wanted to enhance the ability of models to recognize fallacies but data was very scarce.
- They don't use the LLM(GPT-3.5) to classify, instead they use it to generate synthetic examples for other smaller models (T5) to train upon.
- Importance: Turns the few-shot problem into a full-shot solution.

METHODOLOGY

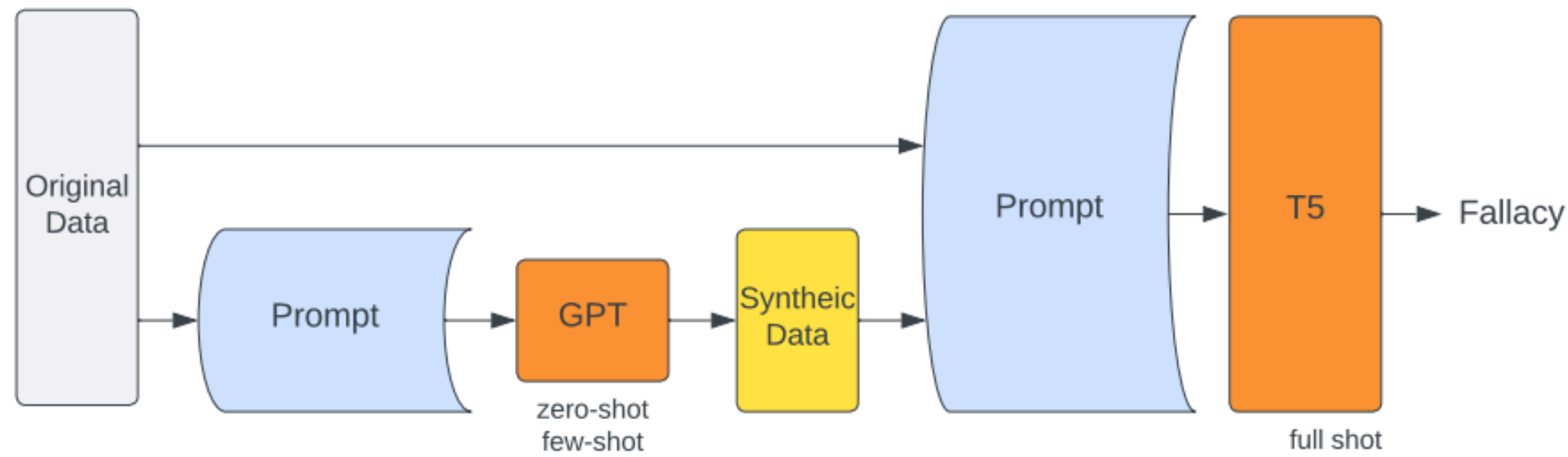


Figure 1: Data augmentation and model training pipeline.

- We take some data from the original one and prompt GPT-3.5 to generate synthetic data.
- Then combine the Original and synthetic data to train the T5 model.

Prompt

Your task is to perform the following actions:

1. Read the sentence that has the fallacy of **Irrelevant Authority** and its provided context. The sentence and context below are both delimited by <>.
2. Generate a similar example of one sentence and its context of the **Irrelevant Authority** fallacy. The context should be a bigger chunk of text that includes the sentence, similar to the provided example sentence and context below.

Your generated output should be in JSON format with the following keys: `generated_sentence`, `generated_context`.

Sentence: <Carlson cited Dr. Robert Epstein who has said, in Carlson's words, "Google alone could determine the outcome of almost any election just by altering its search selections and we would never know it.">

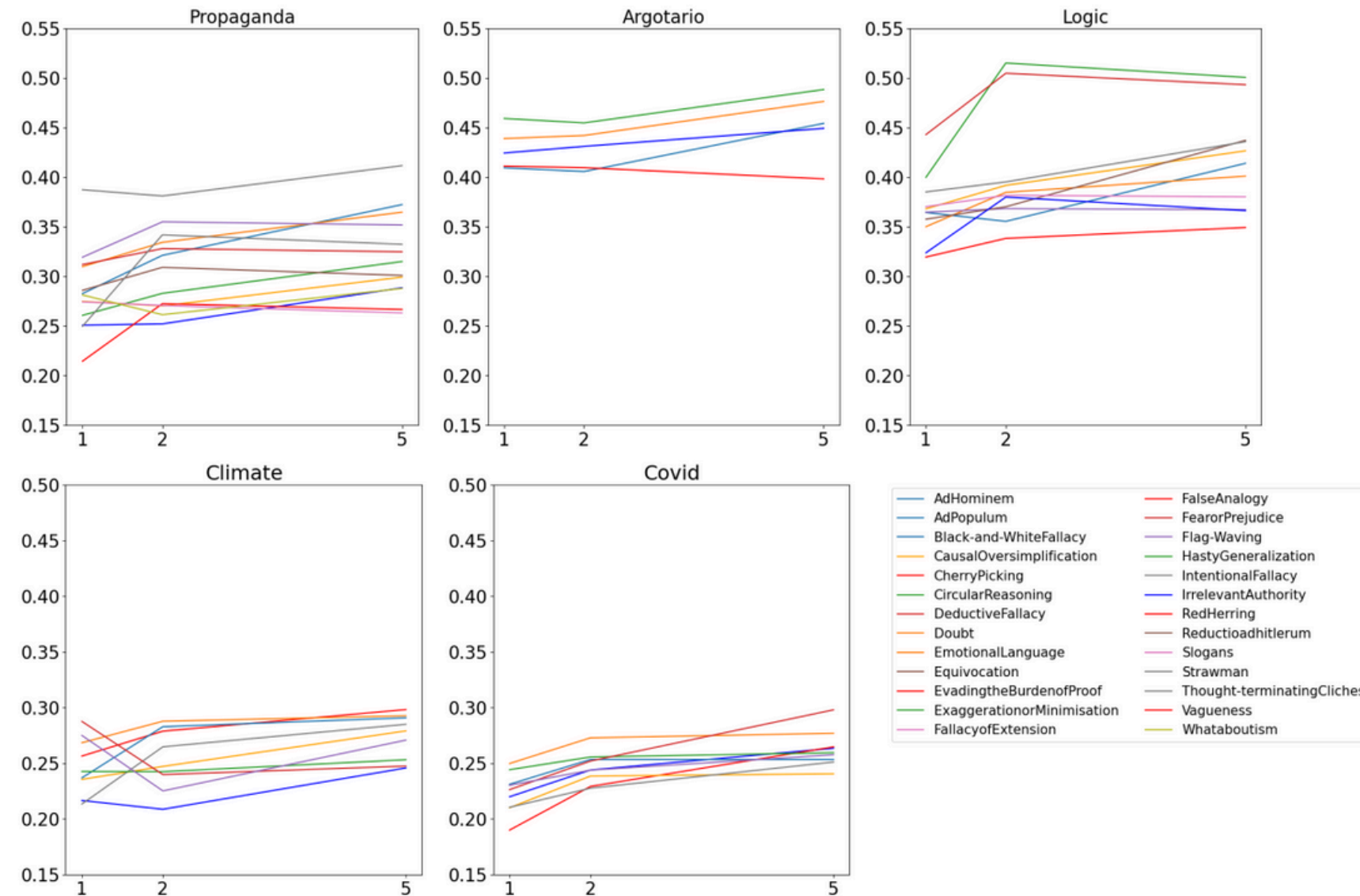
Context: <Carlson cited Dr. Robert Epstein who has said, in Carlson's words, "Google alone could determine the outcome of almost any election just by altering its search selections and we would never know it.". Dr. Robert Epstein is a social scientist and an expert on Google. >

Generated Output

```
{  
  
  'generated_sentence': 'As a dog groomer, I can tell you  
that the earth is flat.',  
  
  'generated_context': 'While most people accept that the  
earth is round, one dog groomer is convinced otherwise.  
As a dog groomer, I can tell you that the earth is flat.  
Dogs are great navigators, and they always sense when  
the ground is flat or sloping.'  
}
```

- Example of a one-shot context prompt.

FINDINGS



- Average BLEURT similarity score between original and synthetic data (on 5 datasets).
- The lower the score, the better, since model is being creative.
- one-shot prompting better for generating data since giving more examples led to overfitting and model started copying answers, resulting in higher similarity scores for more than one shots.

WHERE FEW-SHOT FAILS

- Even when given few-shot examples, the model (GPT-3) failed at physical reasoning tasks. (E.g: “If i put cheese in the fridge, will it melt?”) Since examples only taught the format, not the working of the physical world.
- Guessed randomly on NLI (sentence comparison) tasks even with 32+ examples.

APPLICATIONS

- Translating languages without any “translation training”, so we don't need a separate model for every language pair.
- Content Generation, like realistic articles since model understood style, tone and structure (Human judges could only distinguish between generated articles only 52% of the time).
- Creating data for other models, since LLMs can generate synthetic data to solve the data scarcity problem for rare tasks like fallacy detection.

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THANK YOU